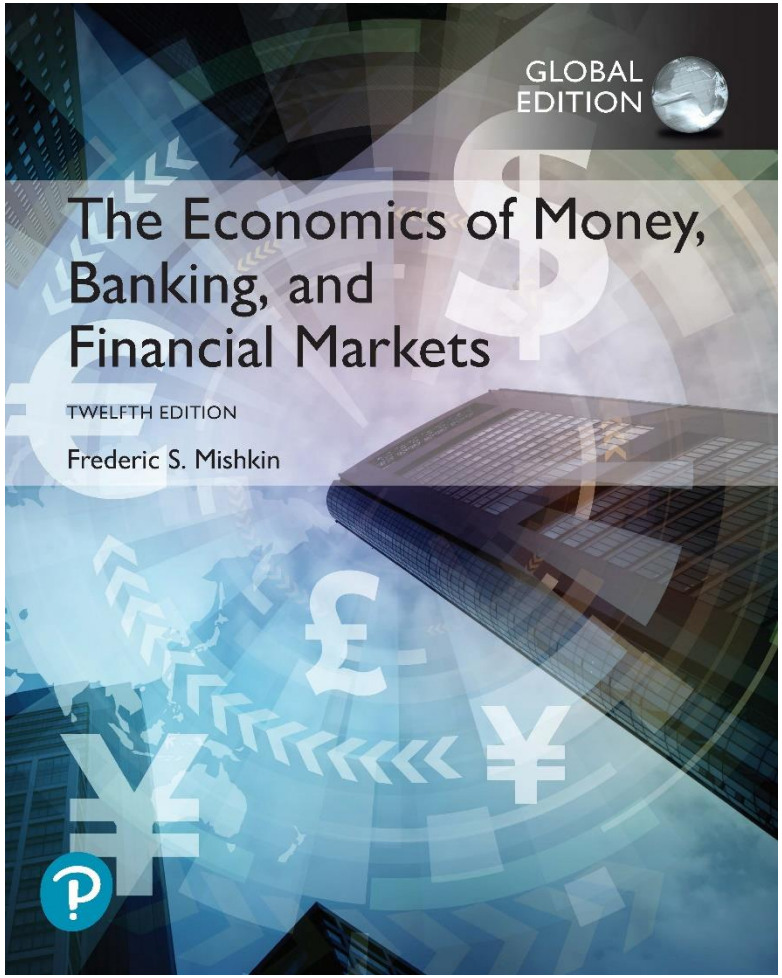
The Economics of Money, Banking, and Financial Markets

Twelfth Edition, Global Edition



Chapter 13

Financial Crises in Emerging Market Economies

Preview

- This chapter applies the asymmetric information theory of financial crises to investigate the cause of frequent and devastating financial crises in emerging market economies
- This analysis is then applied to the events surrounding the financial crises that took place in South Korea and Argentina in recent years and explore why these events caused such devastating contractions of economic activity.

Dynamics of Financial Crisis in Emerging Market Economies (1 of 5)

- Stage one: Initial Phase
  - Path A: Credit Boom and Bust
    - Weak supervision and lack of expertise lead to a lending boom.
    - Domestic banks borrow from foreign banks.
    - Fixed exchange rates give a sense of lower risk.
    - Banks play a more important role in emerging market economies, since securities markets are not well developed yet.
    - At this stage emerging economies may report “sound fundamentals.” For example, fiscal positions may be sound.
    - Credit boom is associated with deteriorating balance of payments.

Key Economic Indicators for Crisis-Stricken Mexico and Thailand

	Mexico				Thailand			
	1992	**1993**	***1994***	**1995**	**1995**	**1996**	***1997***	**1998**
GDP annual growth (in %)	3.5	2.7	4.9	−6.3	8.1	5.6	−2.7	−7.6
Net government lending / borrowing (in % of GDP)	0.1	0.1	−0.6	−3.9	3.0	2.7	−1.7	−6.3
Current account deficit / surplus (in % of GDP)	−8.9	−6.8	−8.1	−0.4	−8.0	−8.0	−2.0	12.5

Source: Adapted from International Monetary Fund. World Economic Outlook Database. www.imfstatistics.org/imf/
Crisis years are indicated in italics.

Dynamics of Financial Crisis in Emerging Market Economies (2 of 5)

- Stage one: Initial Phase
  - Path B: Severe Fiscal Imbalances
    - Governments in need of funds sometimes force banks to buy government debt.
    - When government debt loses value, banks lose and their net worth decreases.
  - Additional factors:
    - Increase in interest rates (from abroad)
    - Asset price
      - Decrease firms' net worth and thus increase adverse selection problems
    - Uncertainty linked to unstable political systems

Dynamics of Financial Crisis in Emerging Market Economies (3 of 5)

- Stage two: Currency Crisis
  - Deterioration of bank balance sheets triggers currency crises:
    - Government cannot raise interest rates (doing so forces banks into insolvency)…
    - … and speculators expect a devaluation.
    - Note difference with advanced economies (where bank failures do not cause currency crisis)
  - Severe fiscal imbalances triggers currency crises:
    - Foreign and domestic investors sell the domestic currency.

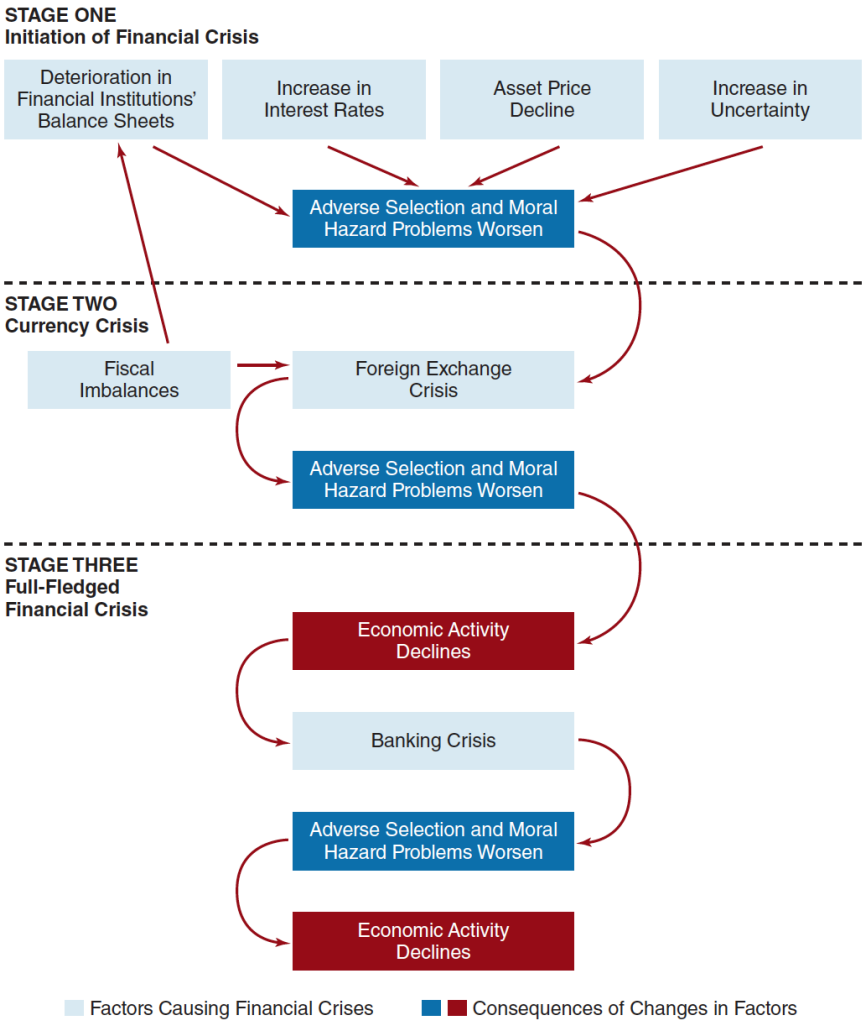
Dynamics of Financial Crisis in Emerging Market Economies (4 of 5)

- Stage three: Full-Fledged Financial Crisis
  - **Currency mismatch:** another difference from advanced economies
    - Arises out of the denomination of many debt contracts in a foreign currency (usually a foreign reserve currency such as the US$)
    - At the same time, however, domestic income denominated in local currency
  - With currency crisis, firms face a higher debt
  - Net worth decreases and averse selection problems increases

Dynamics of Financial Crisis in Emerging Market Economies (5 of 5)

- Stage three: Full-Fledged Financial Crisis
  - Debt burden in terms of domestic currency increases (net worth decreases).
  - Increase in expected and actual inflation reduces firms' cash flow.
  - "Twin crisis"
  - Banks are more likely to fail:
    - Individuals are less likely to payoff debts (value of assets fall)
    - Debt denominated in foreign currency increases (value of liabilities increase)

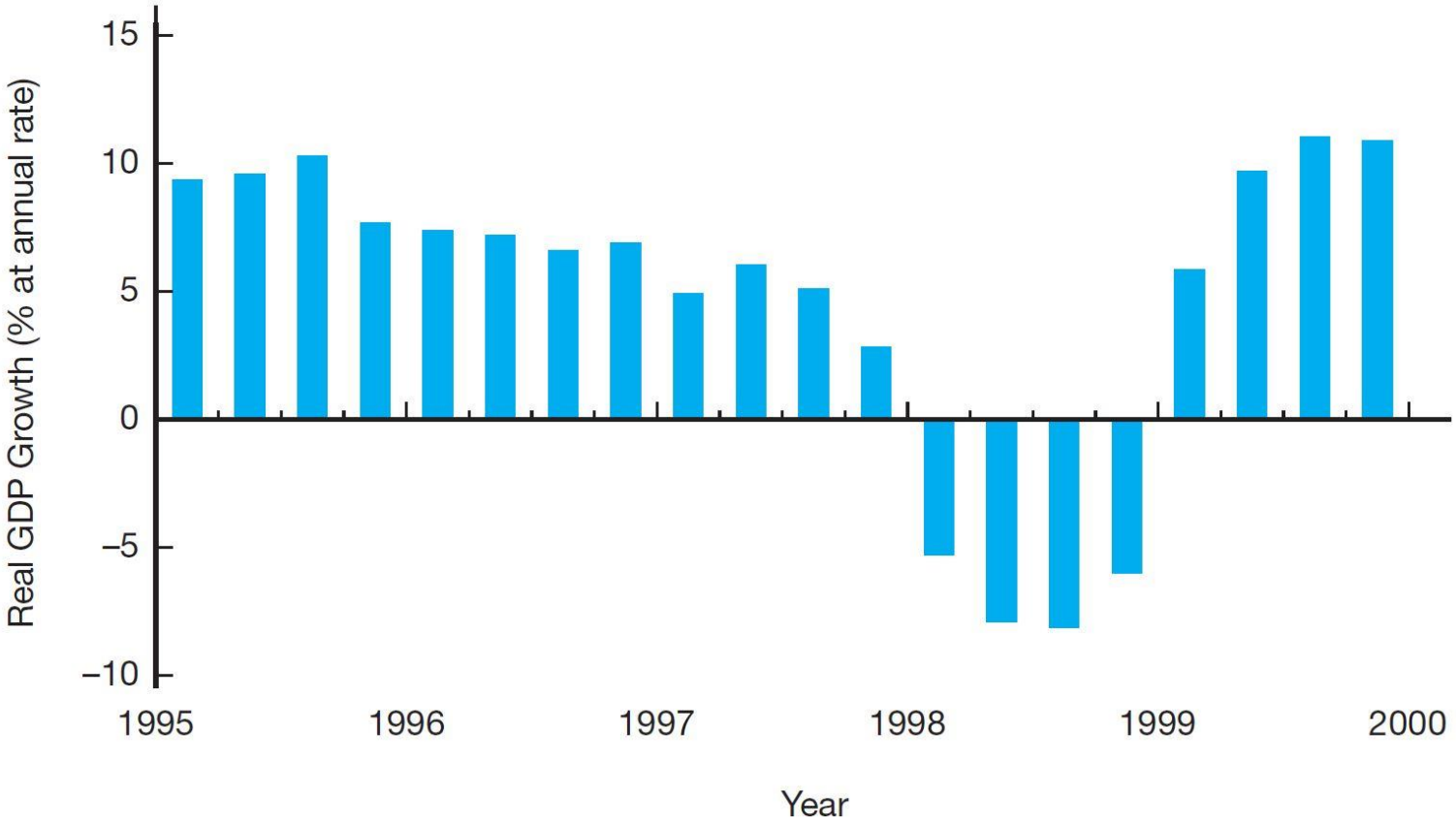
Figure 1 Sequence of Events in Emerging Market Financial Crises



Application: Crisis in South Korea, 1997–98

- Financial liberalization and globalization mismanaged
- Perversion of the financial liberalization and globalization process: chaebols and the South Korean crisis
- Starting point: sound macroeconomic fundamentals
  - Rapid growth, low unemployment, and fiscal surpluses
  - “Asian miracle”

Figure 3 Real GDP Growth, South Korea, 1995–1999



Source: Adapted from International Monetary Fund's *International Financial Statistics*, www.imfstatistics.org/imf/

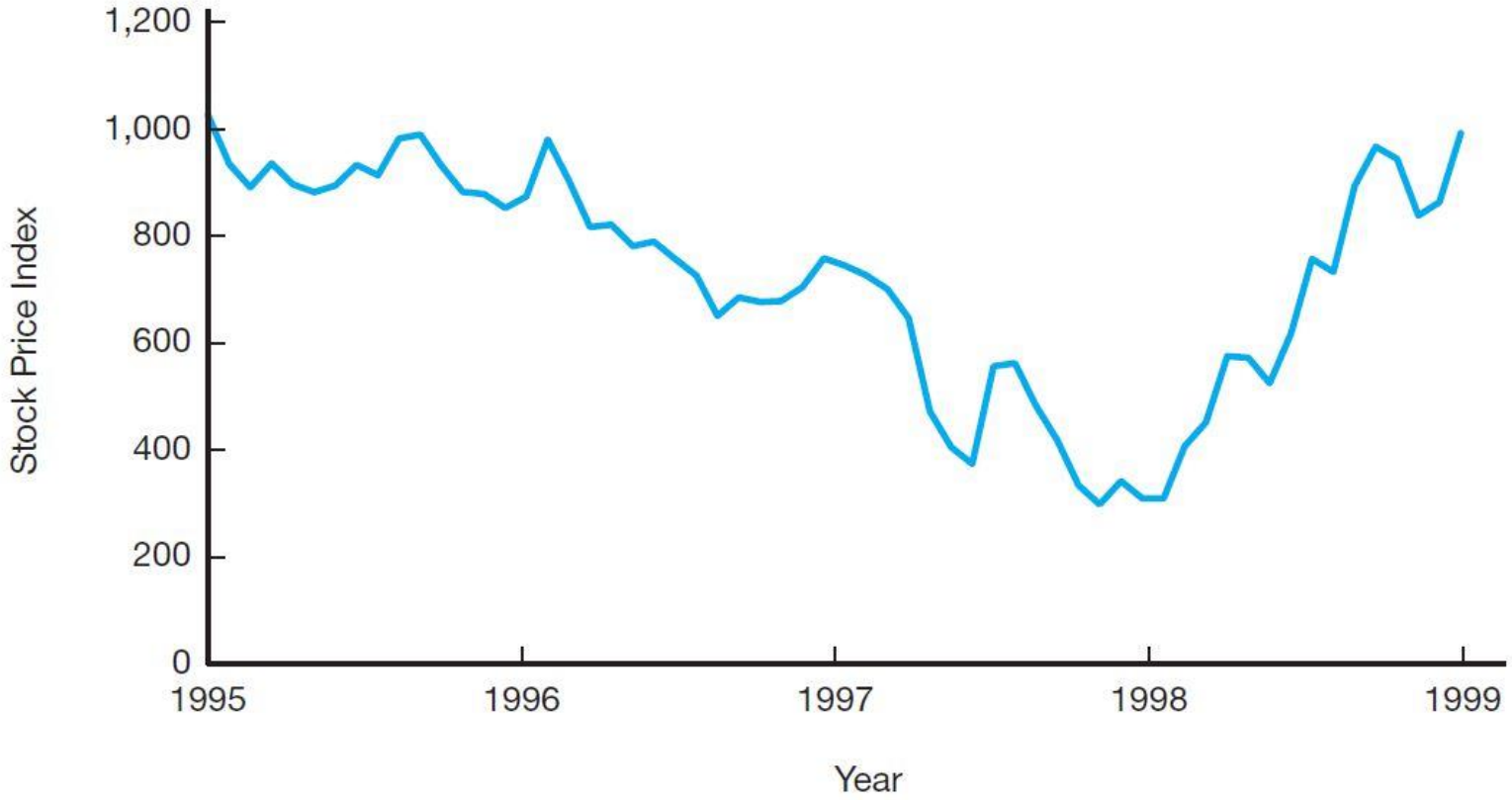
Application: Crisis in South Korea, 1997–98
(2 of 5)

- What happened? Financial liberalization in the 1990s
  - Financial de-regulation
  - Opening up of capital markets
  - Outcome: lending boom fueled by massive foreign borrowing - especially short term
- Perversion of the financial liberalization and globalization process: chaebols and the South Korean crisis
  - Chaebols: large family-owned conglomerates
  - Political ties: too big to fail
  - Leading role in borrowing frenzy: very high leverage (but low profits)
  - Facilitating role played by South Korean merchant banks

Application: Crisis in South Korea, 1997–98
(3 of 5)

- Stock market decline and failure of firms increase uncertainty
  - In early 1997, series of negative shocks to the external position of (already weakened) chaebols
  - First high-profile bankruptcy of one of the largest chaebols in January, followed by 5 more
  - Uncertainty caused by such bankruptcies, and decrease in chaebols' net worth, cause a sharp decline in stock market values
- Adverse selection and moral hazard problems worsen, and the economy contracts

Figure 5 Stock Market Index, South Korea, 1995–1999

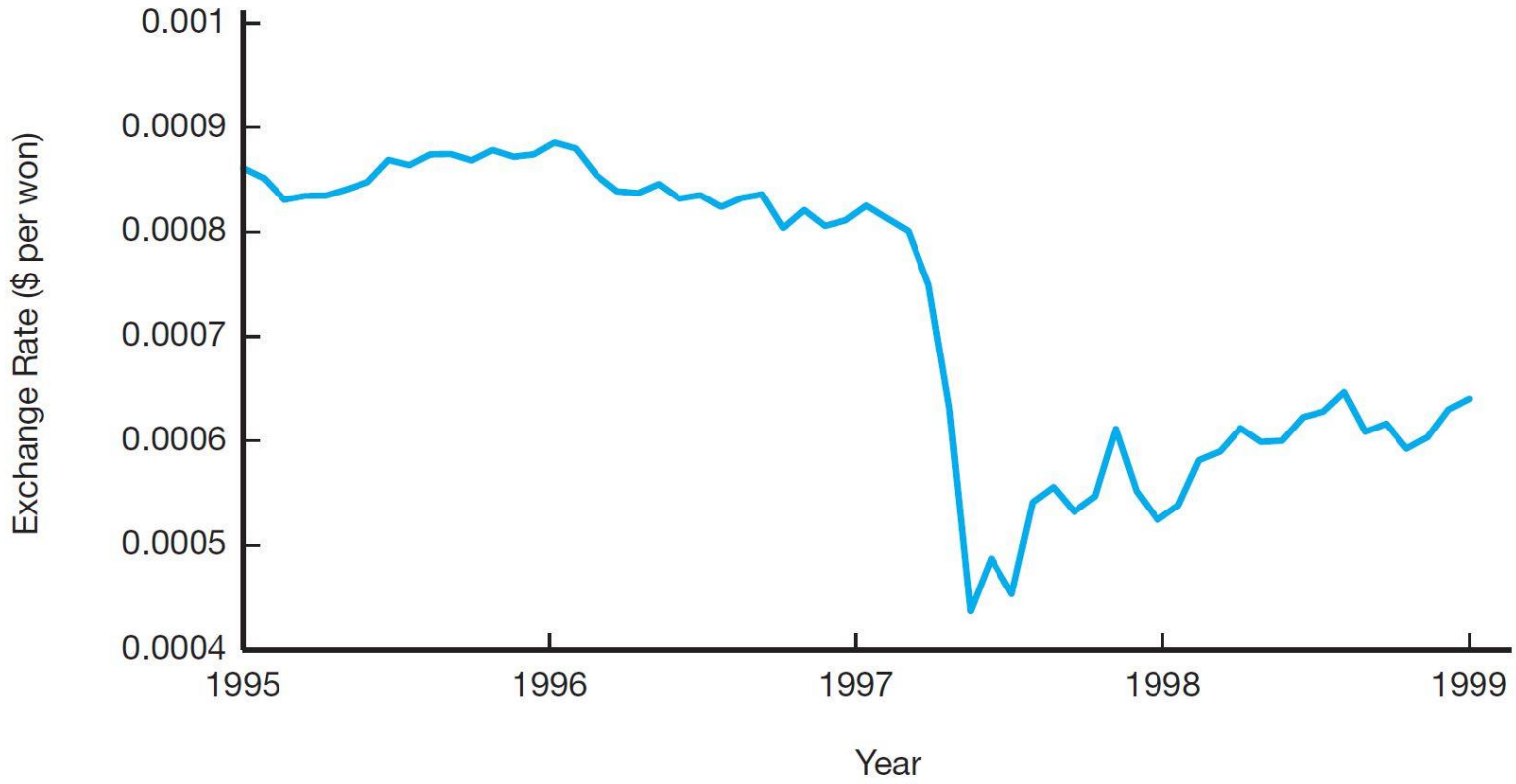


Source: Global Financial Data, available at www.globalfinancialdata.com/.

Application: Crisis in South Korea, 1997–98
(4 of 5)

- Currency crisis ensues
  - Speculative attack against the South Korean currency (won)
  - Fueled by
    - deterioration financial health of chaebols (associated with high leverage)
    - crisis of confidence affecting other Southeast Asian economies
- Final stage: currency crisis triggers full-fledged financial crisis
  - 50% decrease in value of the won increases the value of foreign debt by 50%
  - Increased adverse selection & moral hazard
  - Deterioration of banks' balance sheets
  - GDP declines by 6%

Figure 6 Value of South Korean Currency, 1995–1999



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Application: Crisis in South Korea, 1997–98
(5 of 5)

- Economic impact of the crisis
  - Declining GDP causes a sharp increase in unemployment
  - Inflation increases spurred by increases in import prices
  - Central bank increases interest rates...
  - ...Which worsen the balance sheet of banks and firms
  - Severe socio-economic consequences
- Recovery
  - Aggressive policy by South Korean government to restore financial stability
    - In particular, overhaul of financial regulation
  - Financial markets stabilize in 1998–99.

Global: China and the “Noncrisis” in 1997–1998

- China not affected by the 1997-98 crisis in East Asia. Why?
- Two key factors:
  - Slower, more restricted financial liberalization in China
  - No currency instability
    - Renminbi pegged to the dollar and not convertible for capital account transactions
- Impact of the crisis on China's subsequent economic policies
  - Vindicated the policy of “controlled liberalization”
  - Encouraged the country to accumulate foreign exchange reserves

Application: The Argentine Financial Crisis, 2001–2002 (1 of 6)

- Severe fiscal imbalances
- Adverse selection and moral hazard problems worsen
- Bank panic begins
- Currency crisis ensues
- Currency crisis triggers full-fledged financial crisis
- Recovery begins

Application: The Argentine Financial Crisis, 2001–2002 (2 of 6)

- Severe fiscal imbalances
  - Contrast with Mexico (1995) and East Asian crisis (1997)
  - No lending boom, banks had sounder balance sheets
  - However, growing fiscal imbalances led the government to ask banks to buy public debt
  - Bank balance sheets deteriorate, while investors (foreign and domestic) do not regain trust in Argentina's ability to repay debt

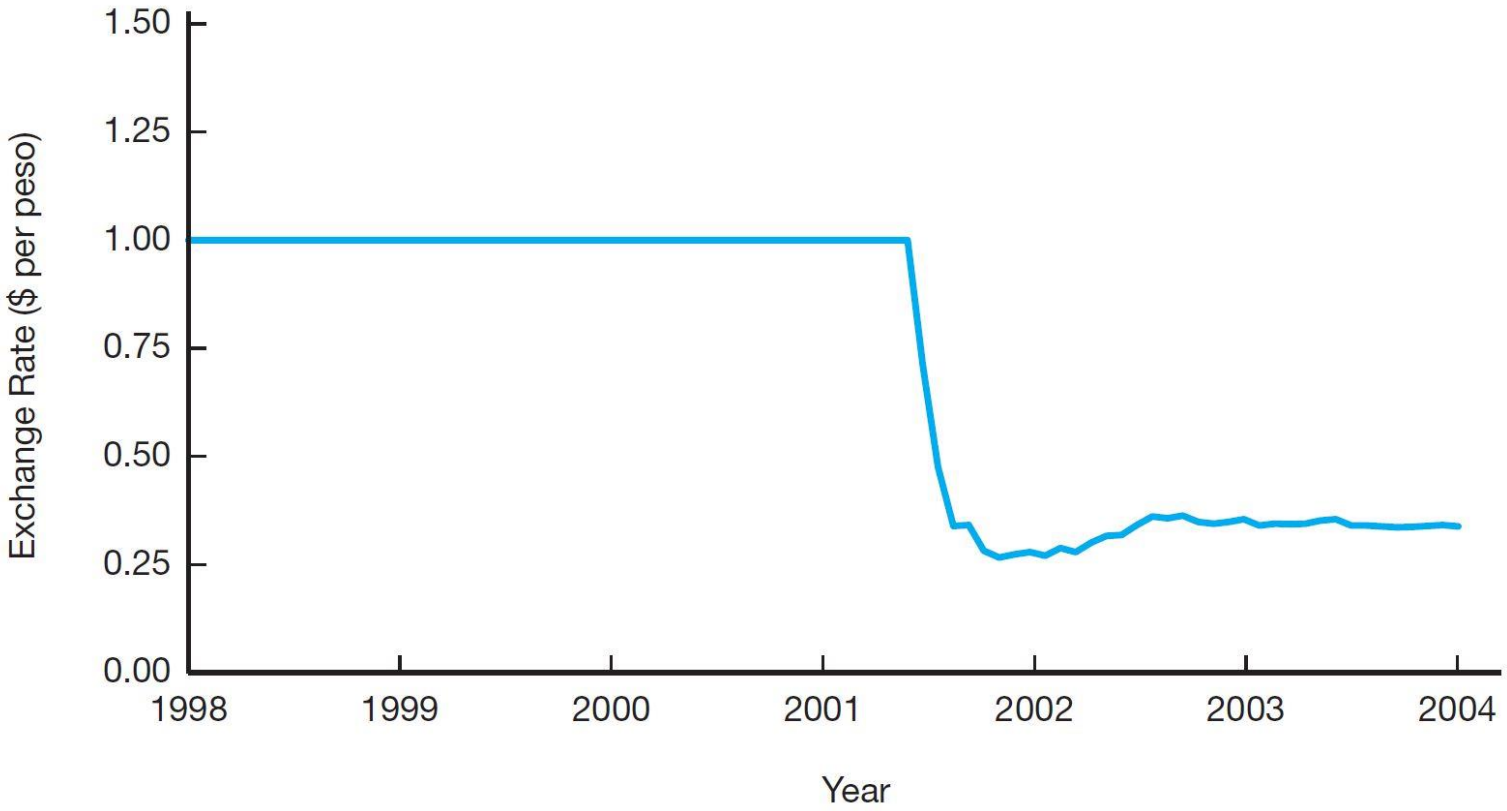
Application: The Argentine Financial Crisis, 2001–2002 (3 of 6)

- Adverse selection and moral hazard problems worsen
  - By 2001, banks start losing deposits and reduce lending
  - Increases in adverse selection and moral hazards
- Bank panic begins
  - Doubts over sustainability of fiscal imbalances grow throughout 2001
  - By October, default becomes real prospect
  - Investors lose trust in banks
  - In November, deposit withdrawals reach US$1 billion/day
  - Restrictions on withdrawals imposed by government fan the flames of social unrest–30 die in violent riots

Application: The Argentine Financial Crisis, 2001–2002 (4 of 6)

- Currency crisis ensues
  - At the time of the bank panic, **currency board** was in place
    - Currency board: Arrangement in which the Argentine government fixed the value of one Argentine peso to one U.S. dollar by agreeing to buy and sell pesos at that exchange rate
  - But bank panic clearly questions the ability of the government to keep interest rates high to preserve the currency board
  - Speculative attacks against the peso multiply
  - In December, 2001, the government announces the suspension of external debt repayments
  - In January 2002, the government announces the end of the currency board

Figure 9 Argentine Peso, 1998–2004

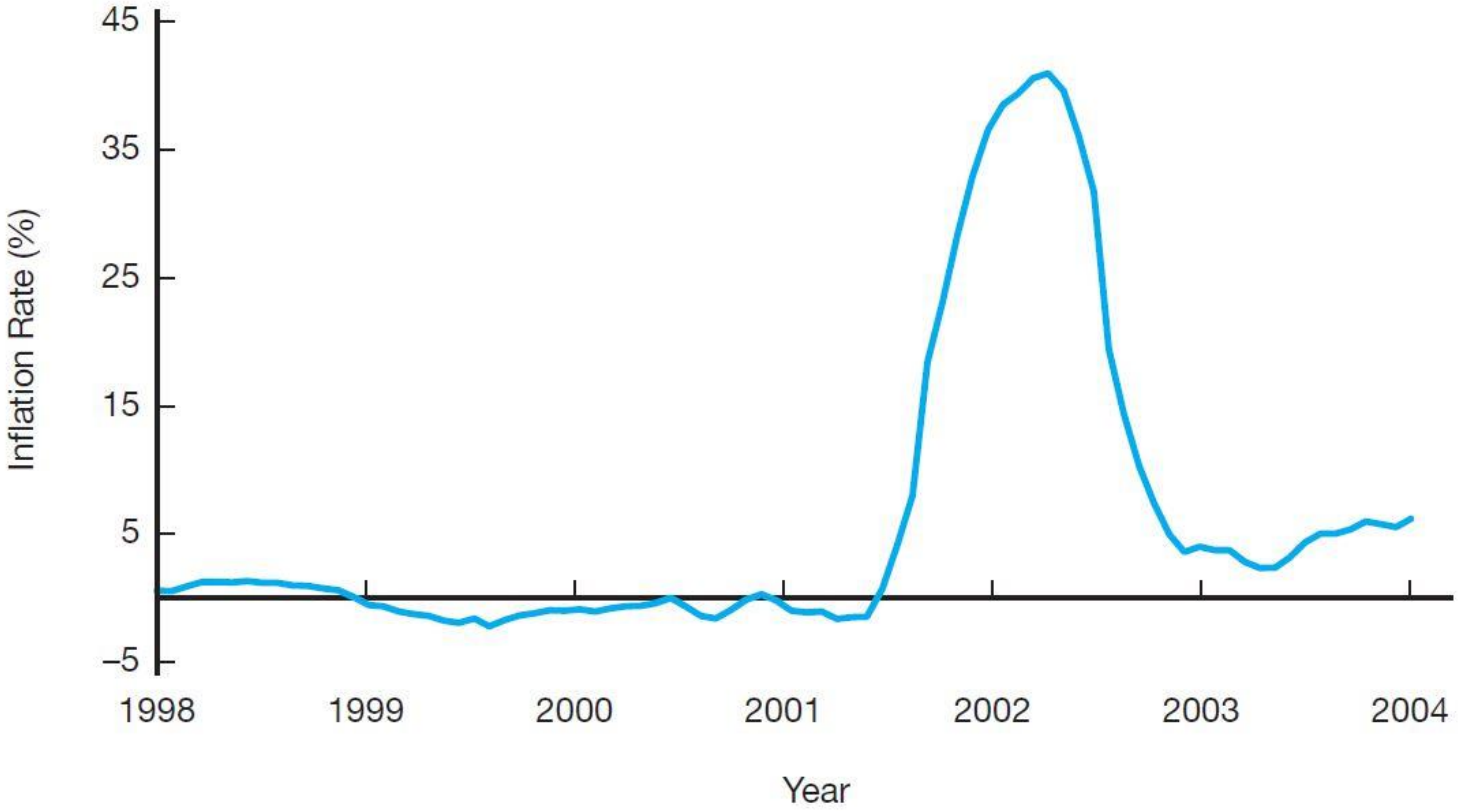


Source: Adapted from International Monetary Fund's *International Financial Statistics*, www.imfstatistics.org/imf/

Application: The Argentine Financial Crisis, 2001–2002 (5 of 6)

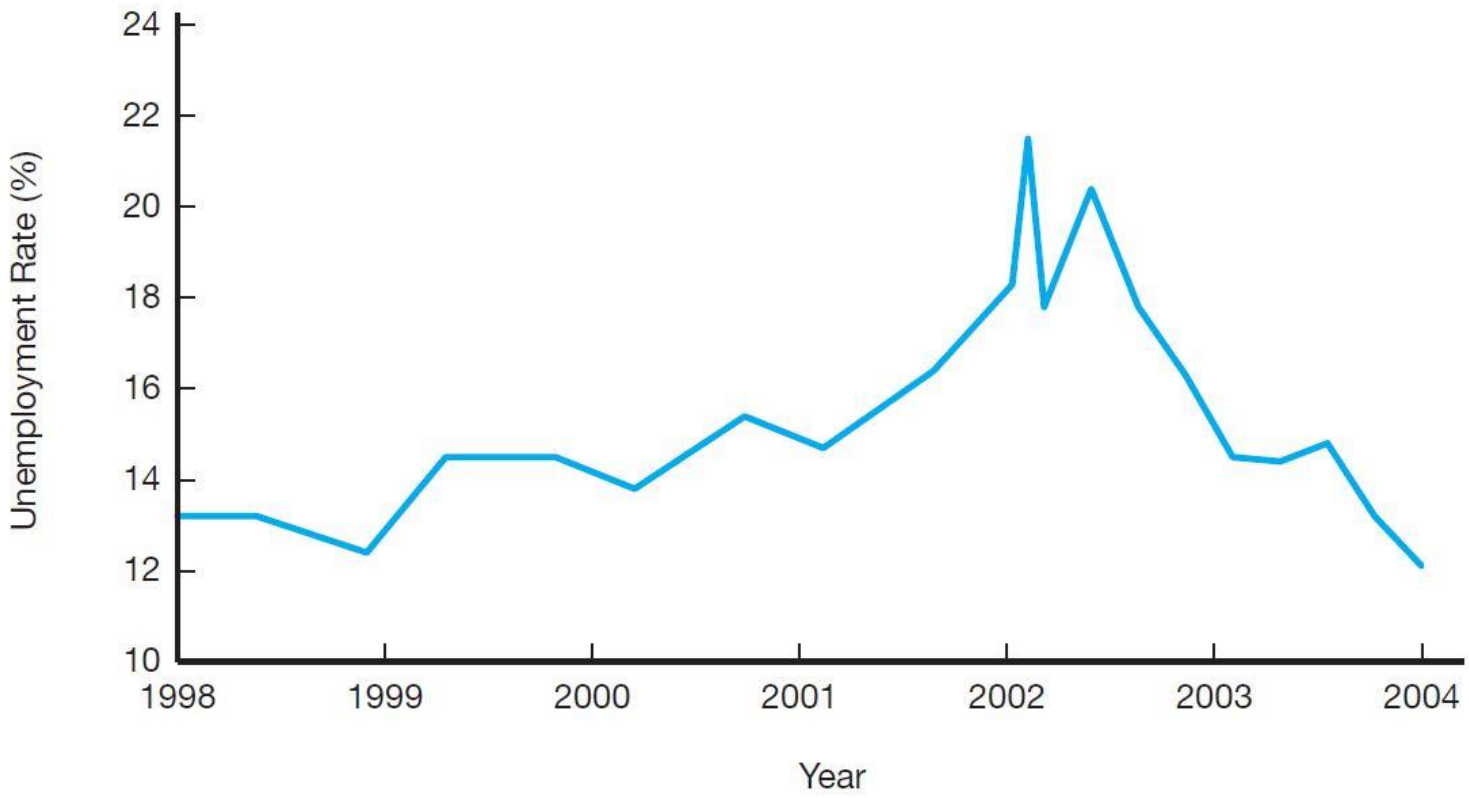
- Currency crisis triggers full-fledged financial crisis
  - Free fall of the peso, stabilizing at 25% of its value by the end of 2001
  - Rapid increase in capital outflows and decline in lending
  - Increase in inflation through higher import prices (like in South Korea), compounded by historically poor inflation performance in the country
  - GDP decreases, dramatic increases in unemployment and poverty
    - Worst crisis in Argentina's history

Figure 10 Inflation in Argentina, 1998–2004



Source: Adapted from International Monetary Fund's *International Financial Statistics*, www.imfstatistics.org/imf/

Figure 12 Unemployment Rate, Argentina, 1998–2004



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Application: The Argentine Financial Crisis, 2001–2002 (6 of 6)

- Recovery begins
  - Financial recovery in 2002 together with boom in demand for Argentinian exports
  - Growth resumes at fast pace in 2003: +10% annual rate

Global: When an Advanced Economy Is Like an Emerging Market Economy: The Icelandic Financial Crisis of 2008

- The financial crisis and economic contraction in Iceland that started in 2008 followed the script of a financial crisis in an emerging market economy, even though Iceland is a wealthy nation.
- Financial liberalization led to rising stock market values and currency mismatch.
- Foreign capital fled the country as a severe recession developed.

Preventing Emerging Market Financial Crises

- Beef up prudential regulation and supervision of banks
- Encourage disclosure and market-based discipline
- Limit currency mismatch
- Sequence financial liberalization